Event Extraction by Answering (Almost) Natural Questions

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As a state-of-the-art, pre-trained language model, BERT continues to gather applause for its ability to extract features with relatively less effort. Its choice as the successor to many of the current non-performant NLP machinery is hence irrefutable. The article resonates with this theme, as we see an event extraction model driven by BERT's question and answering scheme. The merits outweigh most demerits, as evidenced by the improved F1, R, and other metrics employed by the study. Yet this doesn't imply the errors generated during the process are completely random events. The error analysis described in the latter half of this study deliberates on the inefficacies of the Q&A model deployed; entities weren't consistently recognized throughout the experiment. Granted that such complications within sentences are sporadic, this needn't gather much attention other than that occasional cases could cumulatively drive down the evaluation metrics. Nevertheless, such errors provide deeper insights into probable training or data deficiencies within the proposed model. While model breakdowns at complex sentences (failure to capture longer sentences as entities) reveal possibilities that either the final hidden layer in BERT may have masked language modelling training superseding the fine-tuning training component (for event entity recognition) or that the training data supplied to the BERT model lacks representative data (about a population). On similar grounds, failures to capture exact semantic role due to insufficient context also signals improper encoding information being captured in the final layers of BERT. Though seemingly separate issues, both these encounters unanimously suggest that the underlying model performance is capped at the cost of suitable training measures. Indeed, if the attention mechanism is learned on the wrong word dependencies and relationships, the resulting informational encoding cannot be satisfactorily relied on for good predictions. The final contention raised in this study also has a similar background – the contextual word embeddings learned through the model's training may have more profound implications. The assertions presented in the study are valid in that exact matches of rare words are admittedly difficult to find within training examples. However, going by the structural definition of contextual word embeddings, one can then reason how the contextual embedding representation should align with a related word from the training data and hence necessarily find the trigger word in the test set. This predicament may then indicate poor representations being developed upstream of the BERT model, even after sufficient training, signaling how the samples used could be the culprit once more. On a lighter note, we also observe the co-occurrence of peak performances with encoding information from annotation guidelines. Considering how the extrapolation of the same event extraction technique to other possible event scenarios and datasets may demand new annotation guidelines, the readers are inevitably hinted at the presence of a specific 'data dependency'. Suitable pretraining/training may resolve all of this but still warrant a better explanation surrounding such phenomena.